# Internals Know More Than the Model Can Say: Relevance Signals Are Encoded Internally but Attenuated en Route to the Output

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**Abstract**

We study whether large language models (LLMs) "know" more about retrieval relevance than their outputs reveal. Using linear probes on a single frozen forward pass over (query, passage) pairs, we compare four relevance *readers* on the same model, prompt, and answer token: (i) the model's literal output verdict (zero-shot LLM-judge), (ii) a trained probe on the final-layer hidden state (the *output representation*), (iii) a trained probe on the best residual-stream layer (*internal*), and (iv) a probe on the top attention heads. Across six models (3B–8B, LLaMA and Mistral, base and instruct), reading internal activations decodes relevance markedly better than reading the output. With query-clustered bootstrap confidence intervals, the attention-head probe beats the output-representation probe on all four LLaMA models (each $p < 10^{-4}$); after scaling the evaluation to tighten power, the residual-stream advantage over the output representation is individually significant on base models (+0.077, $p$=0.0008; +0.060, $p$=0.004). A causal steering intervention confirms the link is mechanistic, not epiphenomenal: ablating the internal relevance direction at the output layer monotonically collapses the model's verdict (mean $P$(yes) 0.79→0.26) while keeping the verdict discriminative (AUC $\approx$ 0.55 across the sweep). It is the only direction that does both: the output's own direction shifts $P$(yes) but collapses AUC to 0.485 (a non-discriminative logit shift), and a random direction leaves AUC flat without a relevance-aligned verdict shift. Comparing base and instruct pairs, we find alignment improves the output readout (the judge gets stronger) yet does not close the internal–output gap. In short: the relevance signal is encoded internally and *attenuated*, not absent, on the path to the logits.

## 1 Introduction

When an LLM is asked whether a retrieved passage is relevant to a query, its literal answer is often little better than chance. Yet the same forward pass may already contain a far cleaner relevance signal in its internal activations. We ask a focused, mechanistic question: *is relevance better decodable from a model's internals than from its output, and if so, is the internal signal causally connected to the output the model fails to produce?*

This is an interpretability question, not a retrieval-engineering one. We do **not** claim a probe that beats a strong cross-encoder reranker, nor zero-shot cross-domain robustness; prior exploration refuted those framings. Our contribution is the *relative* science of internals vs. output:

- **Existence.** On the same model/prompt/answer-token, a linear probe on attention-head activations decodes (query, passage) relevance substantially better than the model's own output verdict, and better than a trained probe on the output representation (final hidden state).

- **Attribution.** Decomposing the gap into a *training* component (final-layer probe − judge) and a *depth/internal* component (best internal layer − final layer) shows the edge comes substantially from *reading internals*, not merely from *being trained*.
- **Mechanism.** Relevance decodability peaks in the middle of the network and declines toward the output layer. Comparing base vs. instruct models, alignment *improves* the output readout but does not eliminate the internal–output gap.
- **Causality.** A steering intervention that injects the internal relevance direction at the output layer causally controls the verdict; ablation is cleanly monotonic and, unlike the control directions, both moves the verdict and keeps it discriminative—evidence that the output pathway *can* express the signal but normally under-expresses it.

We report query-clustered bootstrap confidence intervals throughout, and are explicit about which claims are individually significant, which rest on a cross-model sign test, and which are architecture-dependent (the attention-head reader is weak on Mistral). Honest delineation of effect strength is central to this paper.

## 2 Related Work

**Probing classifiers** read latent features from frozen representations to test what information is linearly available [Alain and Bengio, 2017, Hewitt and Manning, 2019]; we use them comparatively across depth and against the output. **Logit lens and tuned lens** [nostalgebraist, 2020, Belrose et al., 2023] project intermediate states to the vocabulary, motivating our depth analysis. **Activation steering / representation engineering** [Turner et al., 2023, Zou et al., 2023, Li et al., 2023] edit residual directions to change behavior; we use a difference-of-means steering vector for a causal test. **Calibration and hallucination** work shows models can encode correctness they do not surface [Kadavath et al., 2022, Azaria and Mitchell, 2023, Burns et al., 2022]. Most relevant, Orgad et al. [2024] show LLMs internally represent truthfulness they do not output, and concurrent work finds hallucination is linearly decodable from mid-layer hidden states while output-based detectors stay near chance [Aiersilan, 2026]. We do *not* claim to be first to observe that internals outstrip the output; our contribution is to localize this to a specific, controllable signal—query–passage *relevance*—and to characterize it quantitatively: a depth-resolved attenuation curve, a decomposition separating "reading internals" from "being trained," a causal steering test, and a base-vs-instruct contrast. **Relevance mechanism** work reverse-engineers how neural rankers compute relevance via causal interventions on attention heads [Chen et al., 2024]; trained classifiers are also known to beat prompted LLM judges at relevance [Gienapp et al., 2025]. We differ by studying a generative LLM's internal-vs-output gap on relevance rather than a ranker's internal circuit, and by attributing the gap to depth beyond training. **RAG relevance** estimation usually targets reranking accuracy; we deliberately stay on the interpretability question and do not compete with rerankers.

## 3 Method

**Task and data.** We use MS MARCO v1.1, forming (query, passage, label) triples with relevance labels. Splits are made *by query* (70/15/15) so no query appears in both train and test, preventing

Table 1: Four relevance readers across six models (test AUC, point estimates). *judge*: output verdict, no training; *final*: probe on output representation; *best-resid*: probe on best internal layer; *attn*: top-head probe. The best residual layer and the top heads are selected *on validation* (not on test), consistent with the significance analysis. On every model the best internal reader exceeds both the judge and the output-representation probe. Significance in Tables 2–3.

| Model | $L$ | judge | final | best-resid | attn |
|---|---|---|---|---|---|
| LLaMA-3.2-3B (base) | 28 | 0.535 | 0.592 | 0.628 | **0.685** |
| LLaMA-3.1-8B (base) | 32 | 0.514 | 0.552 | 0.581 | **0.680** |
| Mistral-7B-v0.3 (base) | 32 | 0.531 | 0.601 | **0.621** | 0.603 |
| LLaMA-3.2-3B-Instruct | 28 | 0.597 | 0.606 | 0.633 | **0.725** |
| LLaMA-3.1-8B-Instruct | 32 | 0.603 | 0.583 | 0.628 | **0.746** |
| Mistral-7B-Instruct-v0.3 | 32 | 0.621 | 0.616 | 0.628 | **0.667** |

leakage. We report ROC-AUC on the held-out test set.

**Prompt and extraction.** Every reader sees the identical instruct prompt, `"Document: {p}\nQuestion: {q}\nIs the document relevant to the question? Answer:"`, and all activations are taken at the final *answer token*. Per-head activations are the `o_proj` outputs reshaped to $[\text{heads}, d_{\text{head}}]$; the residual stream is each decoder layer's output.

**Four readers.** (1) *LLM-judge*: the model's own normalized $P(\text{yes})$ vs. $P(\text{no})$ at the answer token, *zero training*. (2) *Output-representation probe*: an $\ell_2$-regularized logistic regression on the final hidden state. (3) *Internal probe*: the same probe on the best residual-stream layer, selected *on validation*. (4) *Attention-head probe*: top-$k$ heads chosen on validation, their activations concatenated into one logistic regression.

**Gap decomposition.** We split the total edge over the judge into train_edge = final − judge (pure training, both at the output) and internal_edge = best_resid − final (pure depth, beyond training). The thesis "internals > output" requires internal readers to exceed *both* the judge and the output-representation probe.

**Significance.** Because the split is by query, the correct resampling unit is the query. We report 95% confidence intervals and one-sided $p$-values from a *query-clustered bootstrap* (5000 resamples) on frozen test predictions; the best layer is chosen on validation to avoid selection-on-test optimism. We additionally run a cross-model sign test on the direction of each edge.

**Causal steering.** At the best internal layer we form a probe-free relevance direction $d = \overline{h}_{\text{rel}} - \overline{h}_{\text{irrel}}$ (difference of class means, unit-normalized). During the forward pass we add $\alpha\|h\|d$ to the *output-layer* residual at the answer token and sweep $\alpha$, reading the judge's $P(\text{yes})$ and the test AUC. We compare against the output layer's own difference-of-means direction and a random unit direction (norm-matched).

## 4 Experiments

**Internals decode relevance better than the output.** Table 1 reports the four readers across six models. On every model the best internal reader exceeds both the zero-shot judge and the trained output-representation probe. The judge itself hovers near chance on base models (AUC 0.51–0.54), confirming the output "says" little.

**Significance.** Table 2 gives query-clustered bootstrap CIs. The *attention-head vs. output-representation* edge is individually significant on all four LLaMA models ($p < 10^{-4}$, effect +0.09 to +0.16). The residual-stream edge (internal_edge) is directionally positive on all six models (sign

Table 2: Query-clustered bootstrap (5000 resamples), 500-query evaluation. Edges with 95% CI and one-sided $p$. **Bold**: $p < 0.05$.

| Model | internal_edge | attn−final | best_resid−judge |
|---|---|---|---|
| LLaMA-3.2-3B | +0.036 [-.05,.12] | +0.094 [.01,.18] | +0.093 [.01,.18] |
| LLaMA-3.1-8B | +0.028 [-.06,.11] | +0.128 [.04,.22] | +0.065 [-.02,.15] |
| Mistral-7B | +0.019 [-.07,.10] | +0.002 [-.08,.09] | +0.089 [.00,.17] |
| LLaMA-3.2-3B-Inst | +0.026 [-.06,.11] | +0.119 [.03,.21] | +0.035 [-.05,.13] |
| LLaMA-3.1-8B-Inst | +0.044 [-.04,.13] | +0.164 [.08,.25] | +0.024 [-.06,.11] |
| Mistral-7B-Inst | +0.011 [-.08,.10] | +0.051 [-.04,.15] | +0.006 [-.08,.09] |
| sign test ($p$) | 6/6 (**.016**) | 6/6 (**.016**) | 6/6 (**.016**) |

Table 3: Scaled (1500-query) re-test of internal_edge vs. the 500-query estimate. **Bold**: $p < 0.05$.

| Model | internal_edge (500q) | internal_edge (1500q) |
|---|---|---|
| LLaMA-3.2-3B (base) | +0.036, $p$=0.19 | +0.077 **[.03,.12],** $p$=0.0008 |
| LLaMA-3.1-8B (base) | +0.028, $p$=0.26 | +0.060 **[.02,.10],** $p$=0.004 |
| LLaMA-3.2-3B-Inst | +0.026, $p$=0.28 | +0.024 [-.02,.07], $p$=0.15 |
| LLaMA-3.1-8B-Inst | +0.044, $p$=0.16 | +0.012 [-.04,.06], $p$=0.32 |

test $p$=0.016) but, at ∼75 test queries, not individually significant—a power problem, not an absent effect.

**Scaling the evaluation confirms the internal edge.** We re-extract four flagship models at 1500 queries (∼225 test queries; sample size is the only change) and re-test (Table 3). The residual-stream edge becomes individually significant on base models (+0.077, $p$=0.0008; +0.060, $p$=0.004): the $p$-values drop from the 0.2 range to the 0.001 range purely from added power. Strikingly, the edge is significant on base models but shrinks and loses significance on instruct models—a mechanistic signal we return to below.

**Depth localization.** Figure 1 (per model in the supplement) shows relevance decodability rising into the mid network and declining toward the output layer; the judge baseline sits well below the peak. For LLaMA the strongest heads cluster in the middle (L10–16); for Mistral single heads carry little signal (early-layer head AUC ≈ 0.50) while the residual stream still encodes relevance—a real architectural difference, not a failure of the thesis.

**Mechanism: base vs. instruct.** Across three base→instruct pairs, alignment shows a consistent triple effect: the judge improves (output "says" more), train_edge falls toward or below zero (training a final-layer probe adds little once aligned), and the internal edge persists (and widens on LLaMA). Combined with Table 3 (the internal edge is significant on base, compressed on instruct), the reading is: *alignment improves the output readout but does not close the internal–output gap*—it does not merely "suppress" the output.

**Causality.** Figure 2 shows the steering sweep. Injecting the internal relevance direction at the output layer drives the verdict monotonically on the ablation arm ($P$(yes) 0.79 → 0.26 as $\alpha$<0) while keeping the verdict discriminative (AUC 0.546 → 0.541, essentially flat). The three directions dissociate cleanly: the internal direction both moves the verdict *and* preserves AUC; the output's own direction moves $P$(yes) strongly (0.01 → 0.99) but *destroys* AUC (→ 0.485, a uniform logit shift); and a random direction leaves AUC flat (≈ 0.545) but produces no relevance-aligned verdict shift. Only the internal direction does both—the diagnostic signature of a causal, relevance-bearing direction rather than a generic bias knob. The additive arm ($\alpha$>0) saturates because the judge

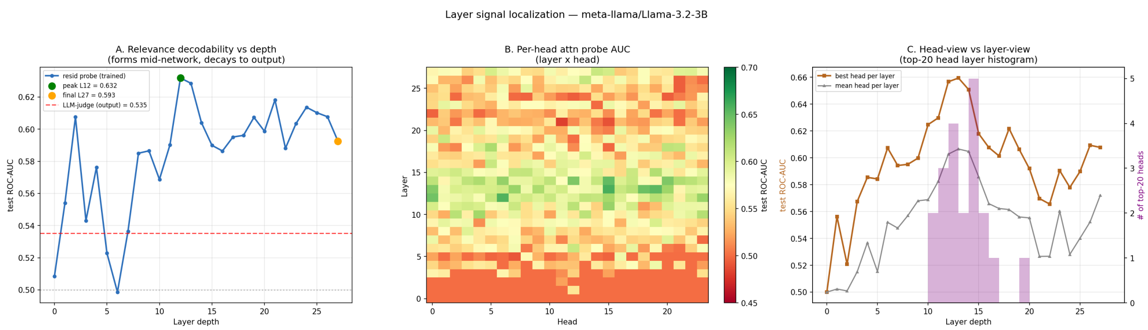


Figure 1: Depth localization (LLaMA-3.2-3B). (a) per-layer residual probe AUC vs. depth with the judge baseline and the peak/output markers; (b) per-head AUC heatmap; (c) head- vs. layer-view and the layer distribution of the top heads. Signal forms mid-network and attenuates toward the output.

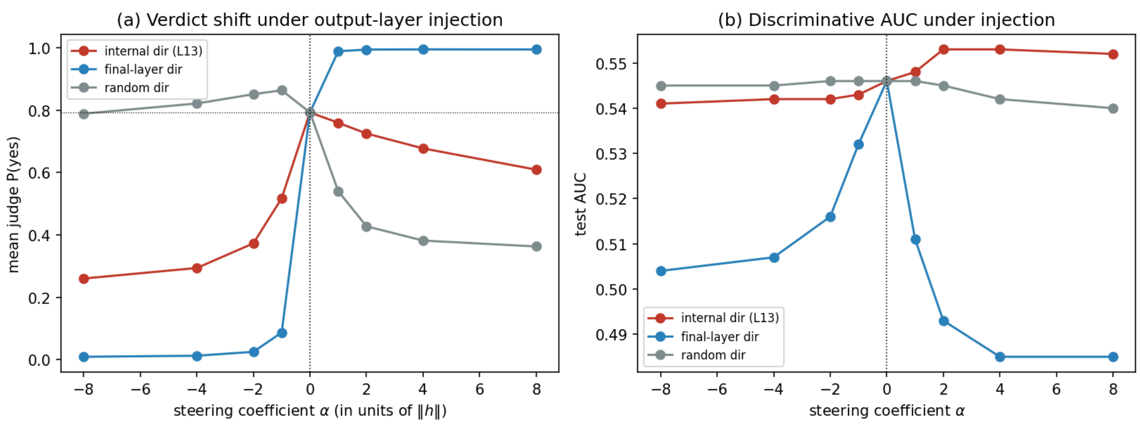


Figure 2: Causal steering at the output layer (LLaMA-3.2-3B). (a) mean judge $P(\text{yes})$ vs. steering coefficient $\alpha$; (b) test AUC. Only the internal direction preserves AUC; the final-layer direction inflates $P(\text{yes})$ while collapsing AUC.

is already yes-biased (baseline 0.79 while only $\sim$22% of test pairs are relevant), so we base the causal claim on the ablation arm and the AUC contrast. This indicates the output pathway *can* express the internal relevance signal but normally under-expresses it: attenuation, not absence. The same protocol replicates across three variants (LLaMA-3.2-3B, LLaMA-3.1-8B base and instruct): the ablation arm collapses $P(\text{yes})$ in every case, and only the internal direction preserves AUC. The collapse is sharpest on the instruct model ($P(\text{yes})$ $0.87 \to 0.003$), mirroring the alignment mechanism above: alignment couples the verdict more tightly to the internal direction without letting the output representation overtake the residual stream.

**A second in-domain dataset.** To check the gap is not specific to MS MARCO, we retrain and re-test the readers in-domain on BeIR FiQA (LLaMA-3.2-3B, 1500 queries, 225 test queries; *not* cross-domain transfer). Negatives are sampled from corpus passages not judged relevant to the query, so FiQA negatives are easier than MS MARCO's in-passage hard negatives; this makes FiQA highly linearly separable and is why its internal_edge is thin (see below). The headline gap is even larger: the judge reaches only AUC $= 0.739$ while the best internal layer reaches 0.993 (best_resid−judge $= +0.253$, CI $[0.224, 0.283]$). FiQA relevance is highly linearly separable, so the output representation itself is near ceiling (AUC $= 0.976$); the internal edge is correspondingly thin but still significant ($+0.017$, CI $[0.010, 0.024]$, $p < 10^{-3}$), and depth still peaks before the output (L14 0.993 vs. L28 0.977). The two datasets are complementary — MS MARCO stresses "reading internals beats reading the output representation," FiQA stresses "a trained read-out far exceeds the zero-shot output judge" — but both show the output judge trailing internal decodability by a wide margin.

**A stronger judge does not close the gap.** A reviewer might attribute the weak judge to a weak prompt. Adding four balanced in-context exemplars (drawn only from train, no leakage) lifts the 3B base judge from AUC $= 0.545$ to only 0.566 — still far below the internal readers (best-resid 0.614, attn 0.682 at the same 1500-query scale). The output bottleneck is attenuation of the signal en route to the logits (Fig. 2), not prompt under-specification.

## 5 Discussion and Limitations

The relevance signal an LLM needs to judge a passage is present in its internals and only partially surfaced at the output; alignment tightens the readout without closing the gap. This connects to calibration/hallucination findings that models "know" more than they assert, localized here to a specific signal and resolved over depth, with a causal handle.

**Limitations.** (1) Probing is correlational; our causal evidence is a steering study replicated on three LLaMA variants (3B, 8B base/instruct), whose additive arm is limited by the judge's yes-bias, so we rely on the ablation arm. (2) We use two in-domain relevance datasets (MS MARCO, FiQA); we make no cross-domain or reranking-accuracy claims, which prior exploration refuted. (3) The attention-head reader is architecture-dependent (weak on Mistral), so we treat the residual-stream layer as the canonical "internal" reader and the head probe as a stronger LLaMA-family instrument. (4) Absolute AUC values are modest (0.5–0.75) on MS MARCO; the contribution is the *relative* internal-vs-output relationship, not a state-of-the-art relevance classifier.

## 6 Conclusion

On the same model, prompt, and token, relevance is more decodable from internal activations than from the output the model produces, the advantage comes substantially from reading depth rather than from training, the signal peaks mid-network and attenuates toward the logits, and a steering ablation shows the internal direction causally underlies the verdict. Internals know more than the model can say.

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